

UrbanPulse-QNX — Requirements Compliance Report

Verification & Traceability against QNX Hackathon Problem Statement SI.No: 49

PROJECT TITLE	PROBLEM STATEMENT SL.NO / DOMAIN
Smart City RTOS Fault & Performance Monitoring Platform (UrbanPulse)	49 / Smart City (Difficulty: Advanced)
TARGET PLATFORM	OVERALL COMPLIANCE STATUS
QNX Neutrino RTOS 8.0 — Dual Raspberry Pi 4 Cluster (aarch64le)	100% COMPLETED (All Mandatory Specs Met)

1. Executive Summary

This report provides a formal evaluation and verification trace of the **UrbanPulse-QNX** distributed monitoring system against the requirements defined in `qnx_Hackthon_Problem_Statement (1).xlsx` (Row 3, Problem Statement #49).

The platform deploys two decoupled nodes over 100 Mbps Ethernet (`genet0`): **Node 1 (Workload Engine)** executing three municipal service threads under fixed-priority rate-monotonic scheduling, and **Node 2 (Supervisor Engine)** ingesting live telemetry, executing statistical percentiles (P95), tracking distributed faults, and driving hardware LED indicators.

2. Requirements Traceability Matrix

#	Excel Requirement	Problem Statement Specification	Implementation in UrbanPulse	Compliance
1	Platform	QNX Raspberry Pi 4/5 Cluster	Dual Raspberry Pi 4 cluster running QNX 8.0 (aarch64le)	COMPLIANT
2	Objective	Monitor task health, CPU utilization, IPC latency and system faults across multiple city services	Tracks Service A (Traffic), B (Water), C (Env); computes P95 latency and moving average CPU	COMPLIANT
3	Node Communication	Ethernet / CAN	TCP/IP stream over port 5555 on <code>genet0</code> Broadcom interface	COMPLIANT
4	System Health Output	GPIO LEDs	BCM2711 direct memory-mapped I/O (Green: 17, Yellow: 27, Red: 22)	COMPLIANT
5	Diagnostic Access	UART / Terminal CLI	Interactive diagnostic shells on both nodes (<code>node1></code> & <code>supervisor></code>)	COMPLIANT
6	Task Starvation	Detect task starvation	Period-aware watchdog checks heartbeat silence > 3x period	COMPLIANT

#	Excel Requirement	Problem Statement Specification	Implementation in UrbanPulse	Compliance
7	Deadline Misses	Detect deadline misses	Hardware timer compares cycle duration against deadlines (80ms/150ms/400ms)	COMPLIANT
8	CPU Overload	Detect CPU overload	Windowed duty cycle evaluated against 70% (Warn) and 85% (Crit) thresholds	COMPLIANT
9	IPC Delays	Detect IPC delays	Shared-memory queue records microsecond latency; tracks P95 > 1ms / > 50ms	COMPLIANT
10	Distributed Failures	Service failures across distributed nodes	Remote task mirror, TCP link loss handler, and 3000ms heartbeat timeout	COMPLIANT
11	Monitoring Task (High)	Tasks & Scheduling	<code>monitoring_task_thread</code> executing at Priority 20 (every 100ms)	COMPLIANT
12	Fault Detector (Highest)	Tasks & Scheduling	<code>hal_gpio_poll_thread</code> executing at Priority 25 (Highest)	COMPLIANT
13	Trace Collector	Tasks & Scheduling	Lock-protected in-memory ring buffer logging microsecond task transitions	COMPLIANT
14	Analytics Engine	Tasks & Scheduling	Dual-node in-memory sorting computing Min, Max, Average, and P95 percentiles	COMPLIANT
15	Inter-task Comm.	Shared Memory + TCP	POSIX mutex/cond shared memory channel on Node 1 + TCP telemetry to Node 2	COMPLIANT
16	Timers	Sampling, Heartbeat, Fault Detection	100ms watchdog timer, per-task periodic timers, 3s node loss timeout	COMPLIANT
17	Logging / Gantt	Gantt Timeline, CPU/IPC Metrics, Fault Map	<code>trace</code> prints Gantt execution log; <code>cpu / ipc / faults</code> print live analytics	COMPLIANT
18	Interface	CLI Mandatory	Full command-driven terminal interface implemented on both nodes	COMPLIANT

3. Architecture & Priority Hierarchy

UrbanPulse enforces fixed-priority preemptive scheduling using QNX `SCHED_FIFO`:

Node	Thread Name	QNX Priority	Functional Role
Pi 1	hal_gpio_poll_thread	25 (Highest)	Hardware level-based button poller (20ms debounce)
Pi 1	monitoring_task_thread	20 (High)	Heartbeat staleness & deadline watchdog (100ms interval)
Pi 1	service_a_thread	15	Adaptive Traffic Control (Period: 100ms, Deadline: 80ms)
Pi 1	service_b_thread	14	Water Grid Pressure Control (Period: 200ms, Deadline: 150ms)
Pi 1	service_c_thread	13	Environmental Quality Filtering (Period: 500ms, Deadline: 400ms)
Pi 1	telemetry_sender_thread	10	TCP telemetry client streaming binary JSON packets (500ms)
Pi 1	cli_thread	5	Interactive diagnostic shell (node1>)
Pi 2	hal_gpio_poll_thread	25 (Highest)	Supervisor physical button poller & tri-color LED driver
Pi 2	supervisor_monitor_thread	20 (High)	Local Supervisor CPU & P95 analytics watchdog (500ms)
Pi 2	telemetry_server_thread	10	TCP ingestion server listening on port 5555 & timeout detector
Pi 2	cli_thread	5	Interactive supervisor diagnostic shell (supervisor>)

4. Live CLI Verification Guide

Inspectors and evaluators can verify compliance live using the following commands:

```
# 1. View Dual-Node Task and Thread Registry
supervisor> tasks
===== TASK & THREAD REGISTRY =====
--- NODE 2 (Supervisor Local Threads - Pi 2) ---
ID   THREAD NAME           PRIORITY  STATE    ROLE
1    hal_gpio_poll_thread   25        RUNNING  Hardware Button Poller (25)
2    supervisor_monitor_thread 20        RUNNING  Local CPU & P95 Watchdog (20)
3    telemetry_server_thread 10        CONNECTED TCP Ingest Server (10)
4    cli_thread             5         ACTIVE   Diagnostic CLI Shell (5)

--- NODE 1 (Workload Remote Tasks Mirror - Pi 1) ---
ID   NAME                   STATE     EXEC_US  MISS  HB
1    Service_A (Traffic)    RUNNING   1820     0     142
2    Service_B (Water)      RUNNING   3410     0     71
3    Service_C (Env)        RUNNING   890      0     28
=====

# 2. View Real-Time Gantt Execution Trace
node1> trace
+045120 us | Service_A | START | 0 us
+046940 us | Service_A | END   | 1820 us
+046950 us | Service_A | HB    | 0 us
+047100 us | Service_B | START | 0 us
+047120 us | Service_B | IPC   | 180 us

# 3. View Dual-Node Statistical CPU Analytics
supervisor> cpu
--- NODE 2 (Supervisor Local - Pi 2) ---
Current   : 1.8 % | Min: 1.4 % | Max: 2.6 % | Avg: 1.9 % | P95: 2.3 % | Samples: 82

--- NODE 1 (Workload Remote - Pi 1) ---
Current   : 24.3 % | Min: 18.2 % | Max: 32.5 % | Avg: 23.8 % | P95: 28.4 % | Samples: 82
```